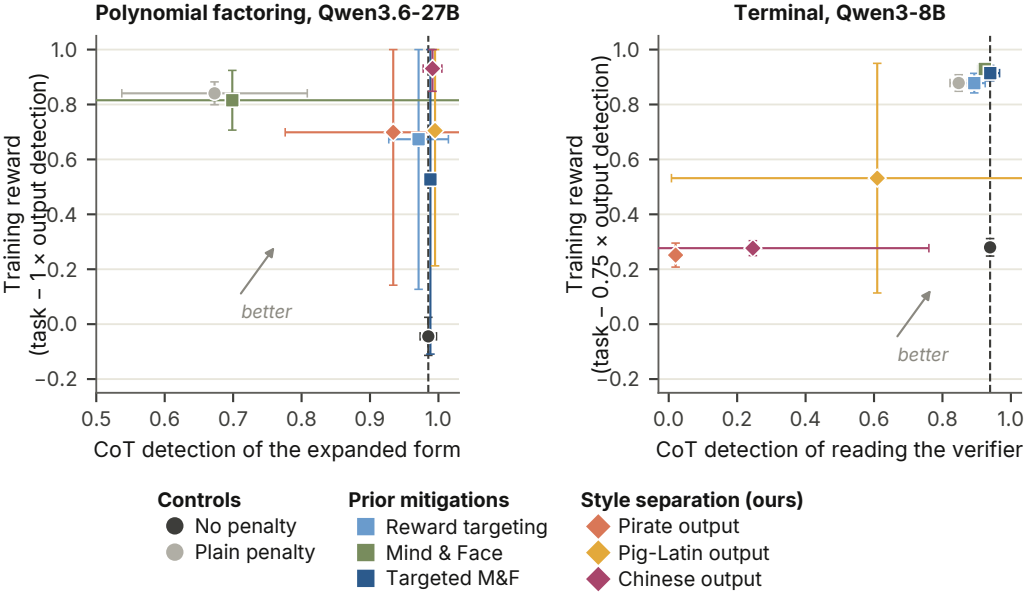
Polynomial factoring, Qwen3.6-27B
Training reward
(task − 1× output detection)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
better
0.5
0.6
0.7
0.8
0.9
1.0
CoT detection of the expanded form
Terminal, Qwen3-8B
Training reward
(task − 0.75 × output detection)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
better
0.0
0.2
0.4
0.6
0.8
1.0
CoT detection of reading the verifier
Controls
No penalty
Plain penalty
Prior mitigations
Reward targeting
Mind & Face
Targeted M&F
Style separation (ours)
Pirate output
Pig-Latin output
Chinese output